

Optimal Control of the Inverted Pendulum

Solving the Algebraic Riccati Equation via Newton's Method

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1 System Model

The inverted pendulum is linearized around the unstable upright equilibrium. The dynamics are represented in state-space form:

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \quad (1)$$

with state vector

$$\mathbf{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ z \\ \dot{z} \end{bmatrix}. \quad (2)$$

Here, θ denotes the pendulum angle and z denotes the cart position.

2 LQR Problem Formulation

The objective is to minimize the infinite-horizon quadratic cost

$$J = \int_0^\infty (\mathbf{x}^T Q \mathbf{x} + \mathbf{u}^T R \mathbf{u}) dt, \quad (3)$$

where $Q \succeq 0$ and $R \succ 0$.

The optimal feedback law is

$$\mathbf{u}^* = -K\mathbf{x}, \quad K = R^{-1}B^T P, \quad (4)$$

where P is the positive-definite solution of the Algebraic Riccati Equation.

3 Algebraic Riccati Equation

Definition 1 (Algebraic Riccati Equation). *The continuous-time Algebraic Riccati Equation (ARE) is*

$$A^T P + P A - P B R^{-1} B^T P + Q = 0. \quad (5)$$

Equation (5) is nonlinear because the unknown matrix P appears quadratically.

Under standard stabilizability and detectability assumptions, there exists a unique positive-definite stabilizing solution P^* .

4 Newton's Method for the ARE

Newton's method solves the nonlinear matrix equation iteratively.

4.1 Residual Function

Define

$$\mathcal{R}(P) = A^T P + P A - P B R^{-1} B^T P + Q. \quad (6)$$

The goal is to find P^* such that

$$\mathcal{R}(P^*) = 0.$$

4.2 Newton Iteration

Given the current iterate P_k , define

$$A_k = A - B R^{-1} B^T P_k. \quad (7)$$

The Newton correction ΔP_k is obtained by solving

$$A_k^T \Delta P_k + \Delta P_k A_k = -\mathcal{R}(P_k). \quad (8)$$

Equation (8) is a Lyapunov equation.

The update is

$$P_{k+1} = P_k + \Delta P_k. \quad (9)$$

4.3 Stopping Criterion

Iterations are terminated when

$$\|P_{k+1} - P_k\|_F < 10^{-8}. \quad (10)$$

Theorem 2 (Quadratic Convergence). *If the initial guess is sufficiently close to the stabilizing solution P^* , Newton's method converges quadratically:*

$$\|P_{k+1} - P^*\|_F \leq c \|P_k - P^*\|_F^2,$$

for some constant $c > 0$.

5 Numerical Considerations

Remark 3 (Symmetry Enforcement). *To avoid loss of symmetry due to numerical round-off,*

$$P_k \leftarrow \frac{P_k + P_k^T}{2}.$$

Remark 4 (Initial Guess). *A common choice is*

$$P_0 = Q.$$

A stabilizing initial guess generally improves convergence.

Remark 5 (Tolerance). *A tolerance of 10^{-8} typically provides an excellent compromise between numerical accuracy and computational effort.*

References

1. F. L. Lewis, D. Vrabie, and V. L. Syrmos, *Optimal Control*, 3rd Edition, Wiley, 2012.
2. B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Dover, 2007.